

EXPLORATORY DATA ANALYSIS (EDA) OF TEEN MENTAL HEALTH DATASET USING PYTHON

I. INTRODUCTION

Exploratory Data Analysis (EDA) is an important step in the data analysis process that helps understand the structure, patterns, trends, and relationships within a dataset. It involves summarizing data using statistical methods and visualizations before applying advanced analytical techniques.

In this project, the Teen Mental Health Dataset was analyzed using Python. Various statistical summaries and graphical visualizations were used to identify trends and factors affecting teenagers' mental health.

II. AIM OF THE PROJECT

- To explore and understand the Teen Mental Health Dataset.
- To perform statistical analysis of the dataset.
- To identify patterns, trends, and relationships among variables.
- To visualize important features using charts and graphs.
- To discover factors influencing stress, anxiety, addiction, and depression levels.

III. SOFTWARE AND LIBRARIES USED

- Google Colab
- Python Programming
- Pandas
- NumPy
- Matplotlib
- Seaborn

IV. DATASET INFORMATION

The Teen Mental Health Dataset contains information related to teenagers' lifestyle, behavior, and mental health conditions.

Dataset Attributes:

- Age
- Gender
- Daily Social Media Hours
- Platform Usage
- Sleep Hours
- Screen Time Before Sleep
- Academic Performance
- Physical Activity
- Social Interaction Level
- Stress Level
- Anxiety Level
- Addiction Level
- Depression Label

V. DATA EXPLORATION PROCESS

Step 1: Importing Dataset

The dataset was loaded into Google Colab using the Pandas library.

Step 2: Dataset Inspection

The shape, column names, data types, and structure of the dataset were examined using functions such as `head()`, `info()`, and `shape()`.

Step 3: Statistical Summary

Descriptive statistics such as mean, median, minimum, maximum, and standard deviation were generated using `describe()`.

Step 4: Missing Value Analysis

The dataset was checked for null values using `isnull().sum()`.

Step 5: Duplicate Record Analysis

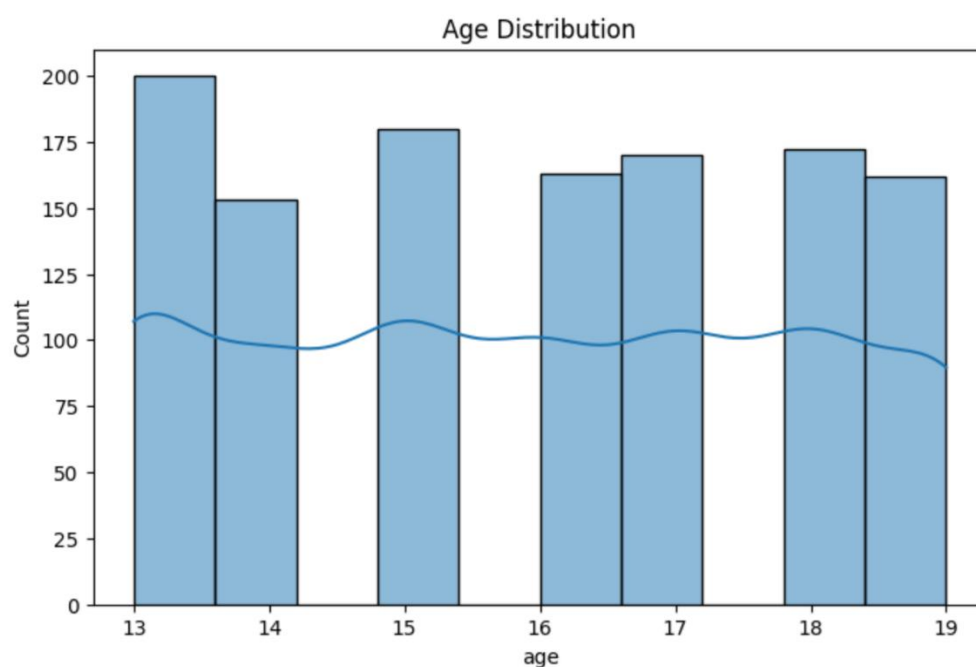
Duplicate records were identified and removed to ensure data quality.

Step 6: Correlation Analysis

Relationships among numerical variables were analyzed using correlation matrices and heatmaps.

VI. DATA VISUALIZATION

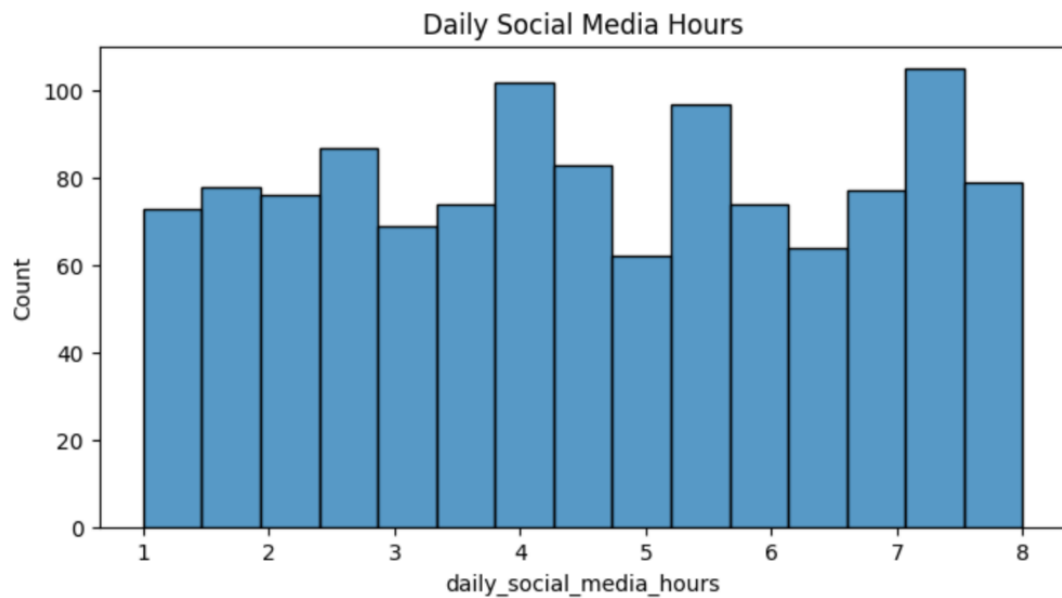
Graph 1: Age Distribution



Observation:

The age distribution graph shows the spread of teenagers across different age groups within the dataset.

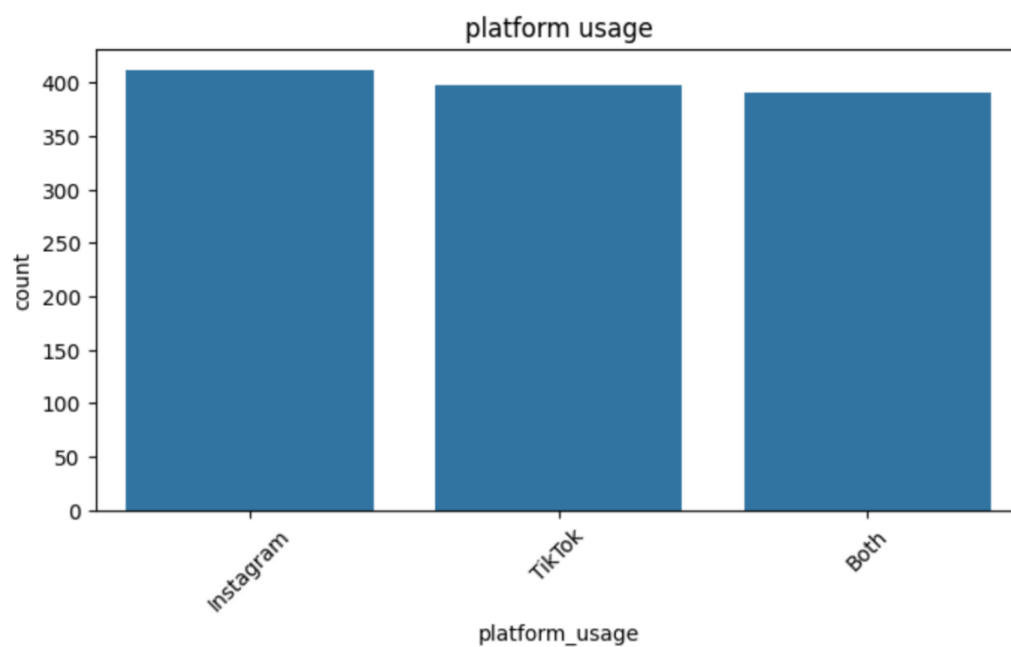
Graph 2: Gender Distribution



Observation:

The dataset contains both male and female participants, providing balanced representation for analysis.

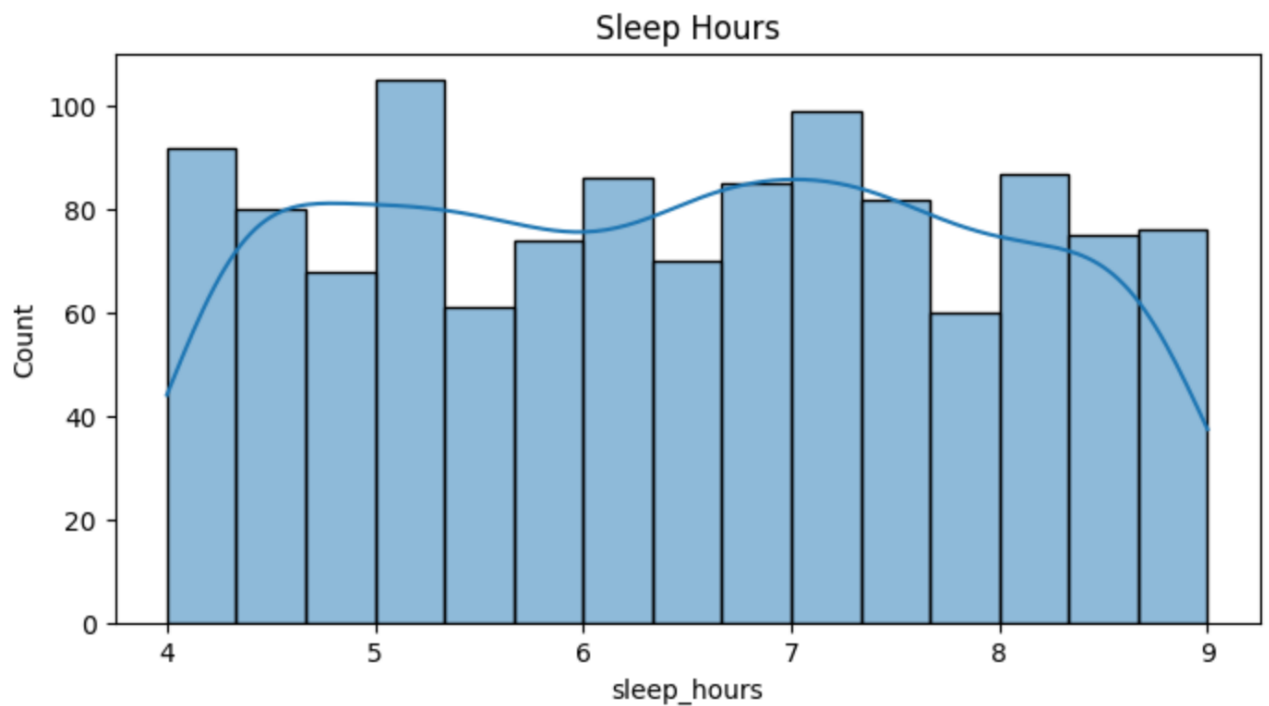
Graph 3: Platform Usage Analysis



Observation:

Different social media platforms show varying levels of popularity among teenagers.

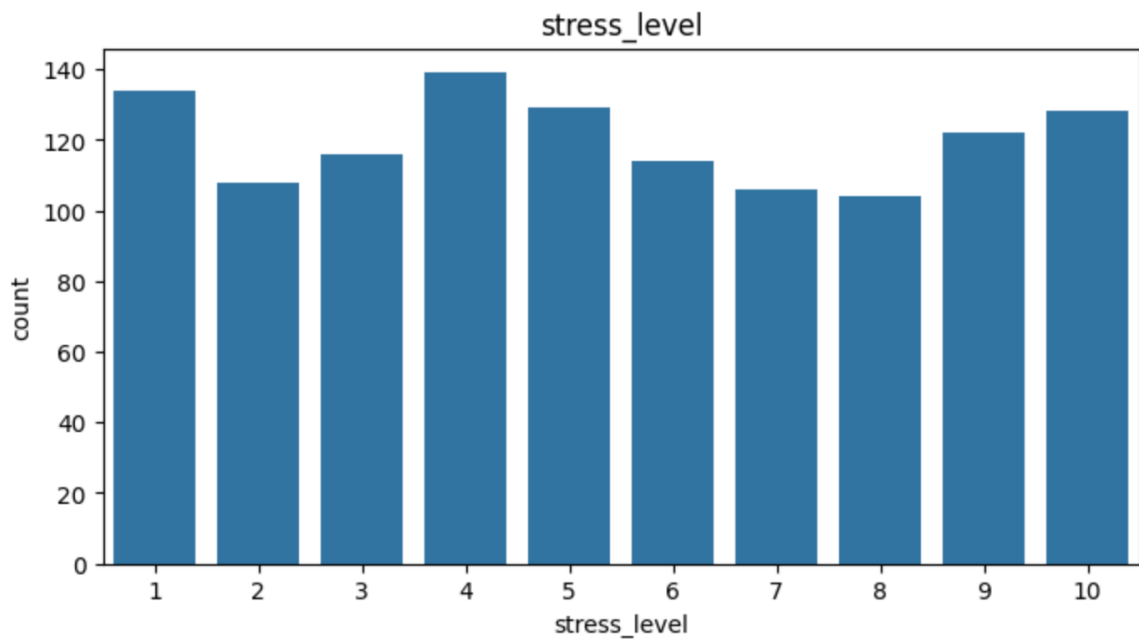
Graph 4: Sleep Hours Distribution



Observation:

Sleep duration varies among participants, with some teenagers receiving fewer hours of sleep than recommended.

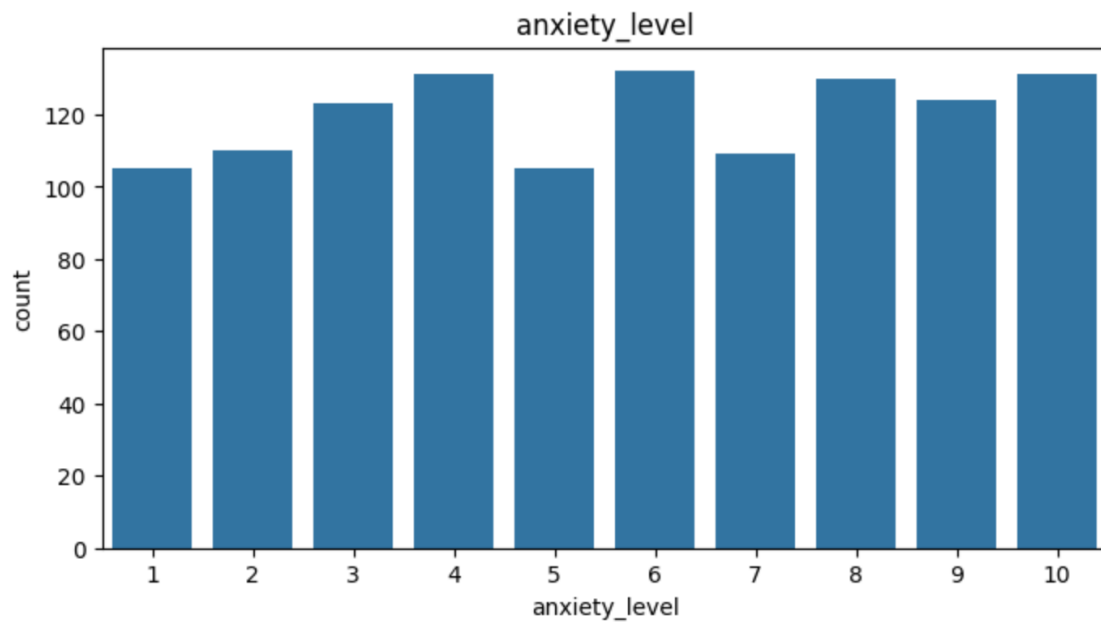
Graph 5: Stress Level Distribution



Observation:

A considerable number of teenagers experience moderate to high stress levels.

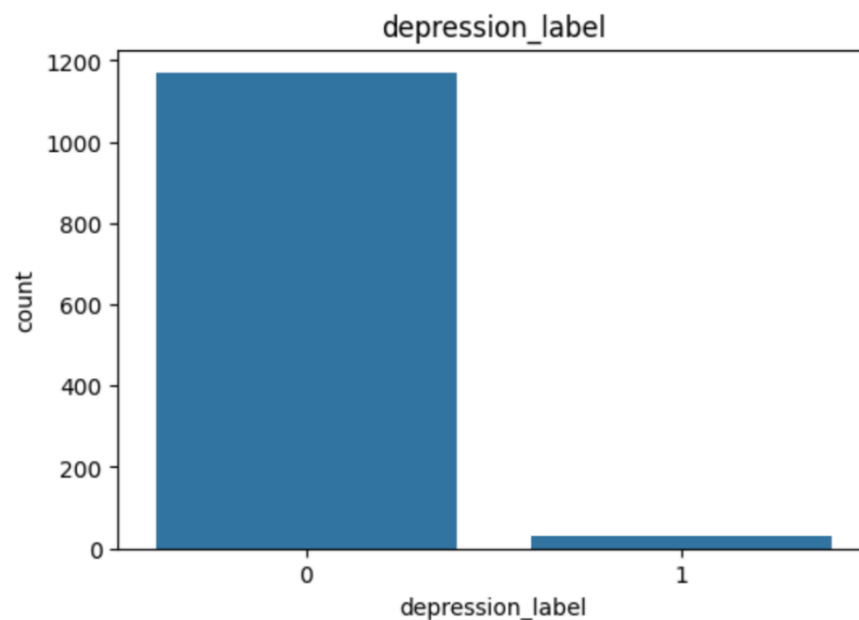
Graph 7: Anxiety Level Distribution



Observation:

The graph indicates varying levels of anxiety among teenagers, highlighting mental health concerns.

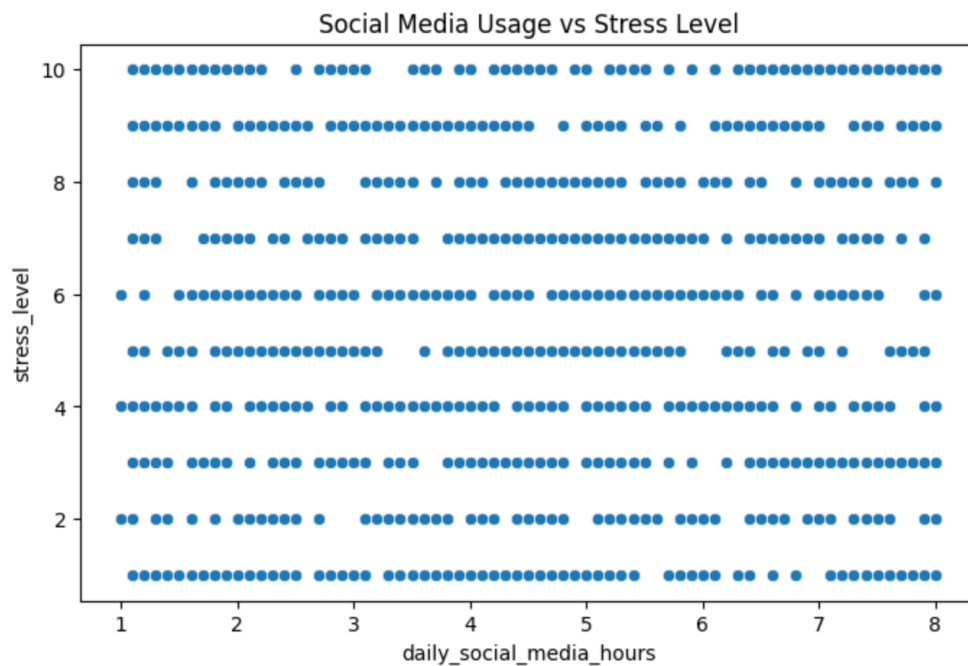
Graph 8: Depression Label Distribution



Observation:

The dataset contains both depressed and non-depressed individuals, enabling comparative analysis.

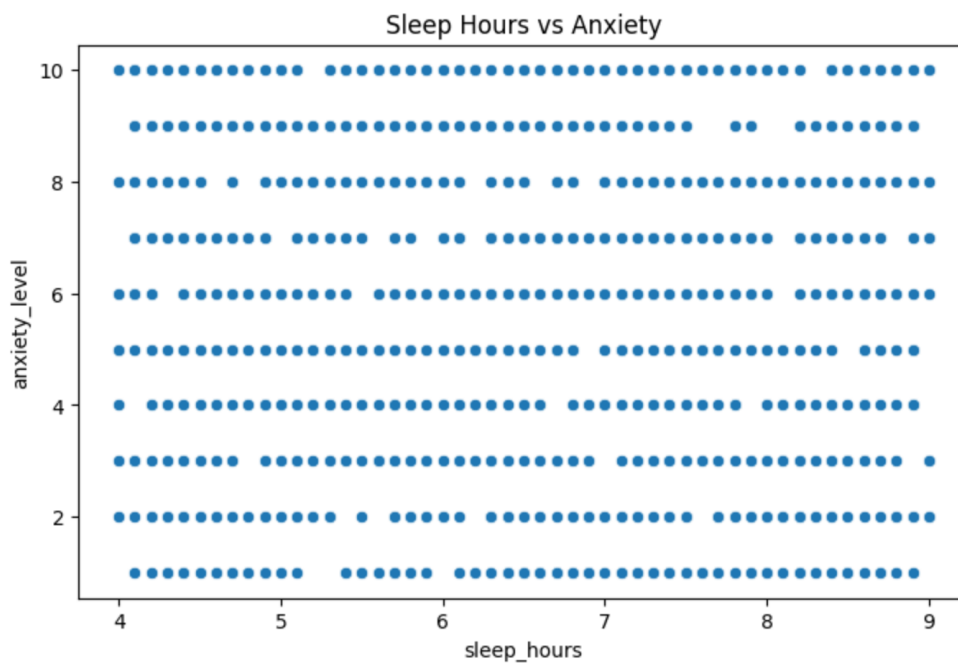
Graph 9: Social Media Usage vs Stress Level



Observation:

Higher social media usage appears to be associated with increased stress levels in some cases.

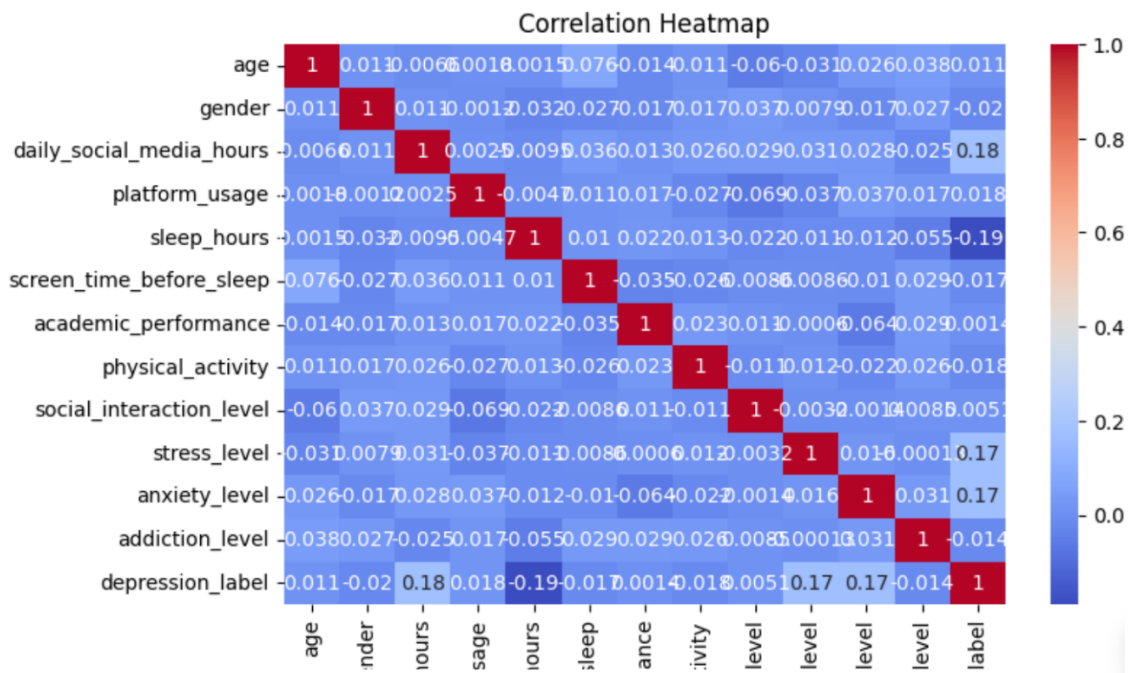
Graph 10: Sleep Hours vs Anxiety Level



Observation:

Teenagers with lower sleep durations tend to exhibit higher anxiety levels.

Graph 11: Correlation Heatmap



Observation:

The heatmap highlights relationships among variables such as stress, anxiety, addiction, sleep hours, and depression.

VII. KEY INSIGHTS

- Social media usage is one of the major factors associated with stress and addiction levels.
- Sleep duration shows a relationship with anxiety and mental well-being.
- Stress and anxiety levels are positively correlated.
- Depression is influenced by multiple lifestyle and behavioral factors.
- Physical activity and social interaction may contribute positively to mental health.

VIII. RESULTS

The exploratory analysis revealed meaningful patterns within the Teen Mental Health Dataset. Statistical summaries and visualizations helped identify important relationships between social media usage, sleep habits, stress levels, anxiety levels, addiction, and depression.

The findings suggest that lifestyle choices significantly influence the mental well-being of teenagers.

IX. CONCLUSION

This project successfully demonstrated the application of Exploratory Data Analysis techniques using Python. Through statistical summaries and visualizations, valuable insights were obtained regarding teen mental health.

The analysis highlighted key factors affecting stress, anxiety, addiction, and depression among teenagers. EDA proved effective in understanding the dataset and preparing it for future machine learning and predictive modeling tasks.

X. REFERENCES

- Teen Mental Health Dataset
- Python Official Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Google Colab Documentation